PR3 Worked Examples: Graphs of Polynomial Functions

Recording companion for the PR3 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What does each factor tell me about the graph? Each factor $(x - a)$ gives a zero at $x = a$, and its exponent is the multiplicity: an even exponent means the graph touches the x-axis and turns around, an odd exponent means it crosses. Plug in $x = 0$ for the y-intercept. Add the exponents for the degree, and look at the sign of the number in front. Together those tell you how the two ends of the graph behave.

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Example 1: Zeros 1

$$f(x)=2(x+4)(x-3)^2$$

Question

What are the zeros and their multiplicities?

- **A.** −3 with multiplicity 2, and 4 with multiplicity 1
- **B.** 3 with multiplicity 2, and −4 with multiplicity 1
- **C.** 3 with multiplicity 1, and −4 with multiplicity 2
- **D.** 2, 3, and −4, each with multiplicity 1

Answer and reasoning

Set each factor equal to 0: $x-3=0$ gives $x=3$, and $x+4=0$ gives $x=-4$. The exponent on each factor is its multiplicity: 2 for $x=3$ and 1 for $x=-4$, since no exponent means 1. The order of the factors does not matter. Reading the signs as written gives −3 and 4, and the 2 in front is a coefficient, not a zero. **3 with multiplicity 2, and −4 with multiplicity 1.**

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Example 2: Zeros 2

$$f(x)=-5(x+2)^2(x-5)^2$$

Question

What are the zeros and their multiplicities?

A. 2 and −5, each with multiplicity 2
B. −2 and 5, each with multiplicity 1
C. −5, −2, and 5, each with multiplicity 2
D. −2 and 5, each with multiplicity 2

Answer and reasoning

$x+2=0$ gives $x=-2$, and $x-5=0$ gives $x=5$. Both factors are squared, so both zeros have multiplicity 2. The −5 in front is the leading coefficient; it changes the height and flips the graph, but it is not a zero. **−2 and 5, each with multiplicity 2.**

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Example 3: Zeros 3

$$f(x)=3(x-4)^2(x+1)$$

Question

What does the graph do at x = 4?

- **A.** It goes straight through the x-axis to the other side.
- **B.** It has a vertical asymptote there.
- **C.** It meets the x-axis and turns back the way it came.
- **D.** It stays above the x-axis without reaching it.

Answer and reasoning

The factor $(x-4)$ is squared, so $x=4$ has even multiplicity. A square is never negative, so $(x-4)^2$ does not change sign as x passes 4: the graph touches the x-axis and turns around. At $x=-1$, multiplicity 1, it crosses instead. Polynomials never have vertical asymptotes. **It touches the x-axis and turns around.**

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Example 4: Zeros 4

$$f(x)=-2(x+5)^2(x-2)$$

Question

Which describes the graph at its zeros?

A. At −5 it turns around at the axis; at 2 it goes through.
B. At −5 it goes through the axis; at 2 it turns around.
C. At 5 it turns around at the axis; at −2 it goes through.
D. It goes through the x-axis at both −5 and 2.

Answer and reasoning

The zeros are -5, with multiplicity 2 because $(x+5)$ is squared, and 2, with multiplicity 1. Even multiplicity: touch and turn around. Odd: cross. Saying it turns at 2 and goes through at −5 swaps them; using 5 and −2 reads the signs as written; going through at both ignores the square. The -2 in front flips the graph upside down, but it does not change touching into crossing. **Touches at −5, crosses at 2.**

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Example 5: Intercept 1

$$f(x)=2(x+4)(x-3)^2$$

Question

What is the y-intercept?

A. (0, 24)
B. (0, −72)
C. (0, 2)
D. (0, 72)

Answer and reasoning

$f(0)=2(0+4)(0-3)^2=2\cdot 4\cdot 9=72$. Square first: $(-3)^2=9$, which is positive. The answer 24 forgets the square, $2\cdot 4\cdot 3$; −72 squares -3 to get -9; and 2 is just the coefficient. ***(0,72)*.**

Example 6: Intercept 2

$$f(x)=-3(x+2)^2(x-5)^2$$

Question

What is the y-intercept?

A. (0, −300)
B. (0, 300)
C. (0, −30)
D. (0, −150)

Answer and reasoning

$f(0)=-3(2)^2(-5)^2=-3\cdot 4\cdot 25=-300$. Both squares are positive, so the sign comes only from the −3. The answer 300 loses that sign, −30 forgets both squares, and −150 squares only one of them. ***(0,−300).***

Example 7: Ends 1

$$f(x)=4(x-2)^2(x+3)$$

Question

Which describes the end behavior?

- **A.** Both ends go up
- **B.** Left end down, right end up
- **C.** Left end up, right end down
- **D.** Both ends go down

Answer and reasoning

The degree is $2+1=3$, which is odd, so the ends go opposite ways. The leading coefficient 4 is positive, so the graph goes up on the right, like $y=x^3$. In CheckIt's words: as x approaches infinity, y approaches positive infinity; as x approaches negative infinity, y approaches negative infinity. **Down on the left, up on the right.**

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Example 8: Ends 2

$$f(x)=-2(x+4)^2(x-1)^2$$

Question

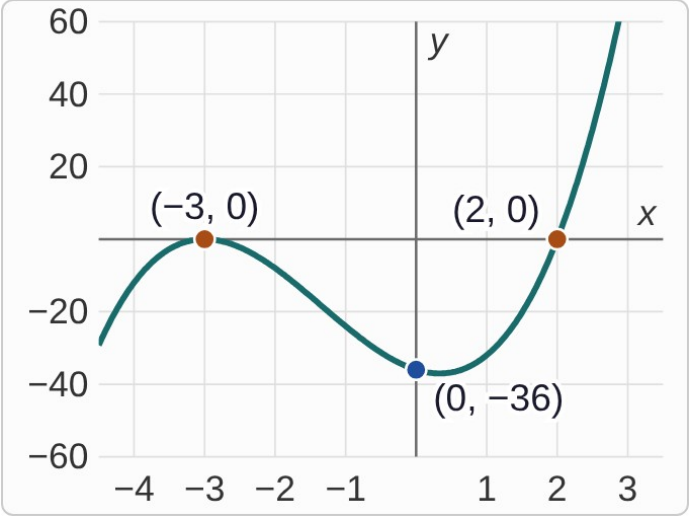
Which describes the end behavior?

A. Both ends go up
B. Left end down, right end up
C. Both ends go down
D. Left end up, right end down

Answer and reasoning

The degree is $2+2=4$, which is even, so both ends go the same way. The leading coefficient -2 is negative, so both go down, like $y=-x^4$. In CheckIt's words: as x approaches infinity and as x approaches negative infinity, y approaches negative infinity. **Down on both the left and the right.**

Example 9: Match 1



Plotted data: A graph on axes with x from about −4.5 to 3.5 and y from −60 to 60. Coming from the lower left, it rises to meet the x-axis at (−3, 0), turns back down without going above the axis, reaches a low point just right of the y-axis, passing through the y-intercept (0, −36), then rises through the x-axis at (2, 0) and keeps rising to the upper right. The two zeros and the y-intercept are labeled.

Question

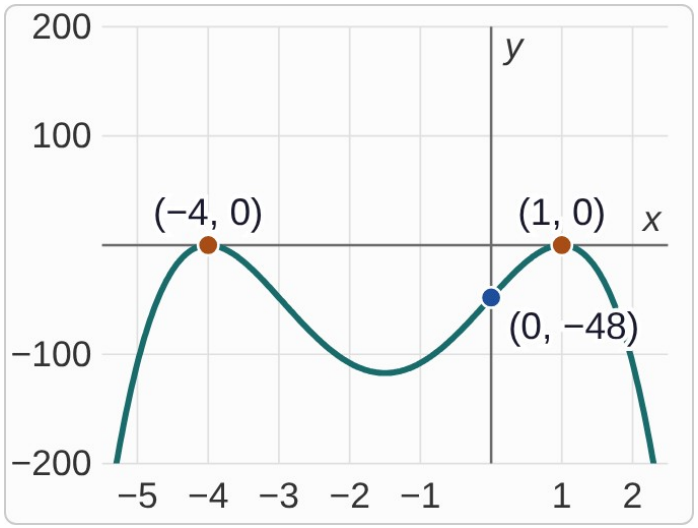
Which formula matches this graph?

- **A.** $f(x)=2(x+3)^2(x-2)$
- **B.** $f(x)=2(x+3)(x-2)^2$
- **C.** $f(x)=-2(x+3)^2(x-2)$
- **D.** $f(x)=2(x-3)^2(x+2)$

Answer and reasoning

The graph turns around at -3, so $(x+3)$ has an even exponent, and it crosses at 2, so $(x-2)$ has an odd one. The ends go down on the left and up on the right: odd degree, positive leading coefficient. Check the y-intercept: $2(3)^2(-2)=-36$. Swapping the exponents would make it cross at −3; the −2 version would flip the ends; and $(x-3)^2(x+2)$ has zeros at 3 and −2. ***f(x)=2(x+3)²(x−2).***

Example 10: Match 2



Plotted data: A graph on axes with x from about −5.5 to 2.5 and y from −200 to 200. Coming up from the lower left, it rises to meet the x-axis at (−4, 0) and turns back down, falls to a low point near x = −1.5, rises through the y-intercept (0, −48) to meet the x-axis again at (1, 0), turns back down, and falls to the lower right. The graph never goes above the x-axis. The two zeros and the y-intercept are labeled.

Question

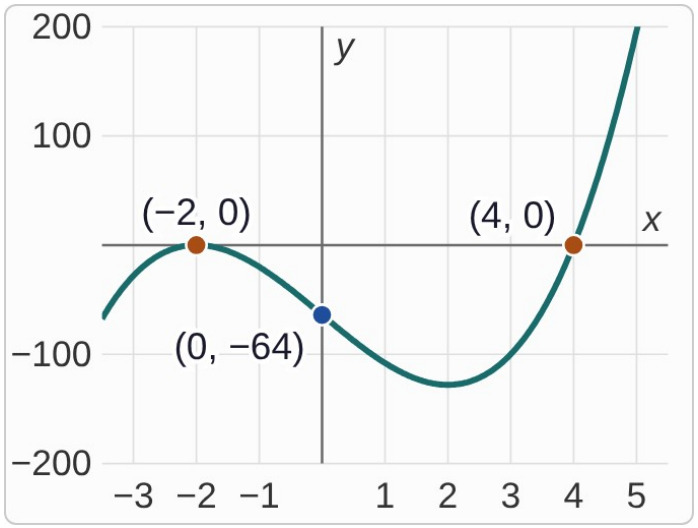
Which formula matches this graph?

- **A.** $f(x)=3(x-1)^2(x+4)^2$
- **B.** $f(x)=-3(x-1)(x+4)^2$
- **C.** $f(x)=-3(x+1)^2(x-4)^2$
- **D.** $f(x)=-3(x-1)^2(x+4)^2$

Answer and reasoning

It turns around at both zeros, so both exponents are even. Both ends go down: even degree, negative leading coefficient. Zeros at 1 and −4 mean factors $(x-1)$ and $(x+4)$. Check: $-3(-1)^2(4)^2=-48$. The positive version would point both ends up, the version with $(x-1)$ to the first power would cross at 1, and $(x+1)^2(x-4)^2$ has zeros at −1 and 4. **$f(x)=-3(x-1)^2(x+4)^2$.**

Example 11: Match 3



Plotted data: A graph on axes with x from about −3.5 to 5.5 and y from −200 to 200. Coming from the lower left, it rises to meet the x-axis at (−2, 0), turns back down without going above the axis, passes through the y-intercept (0, −64), reaches a low point near x = 2, then rises through the x-axis at (4, 0) and keeps rising to the upper right. The two zeros and the y-intercept are labeled.

Question

Which formula matches this graph? Use the y-intercept.

- **A.** $f(x)=2(x+2)^2(x-4)$
- **B.** $f(x)=4(x+2)^2(x-4)$
- **C.** $f(x)=4(x+2)(x-4)^2$
- **D.** $f(x)=-4(x+2)^2(x-4)$

Answer and reasoning

Turns at −2 and crosses at 4, with ends down-left and up-right, so the form is $k(x+2)^2(x-4)$ with k positive. The shape alone cannot tell 2 from 4; the y-intercept can: $k(2)^2(-4)=-16k=-64$, so $k=4$. With $k=2$ the intercept would be $(0,-32)$. The other two choices cross at the wrong zero or flip the ends. ***$f(x)=4(x+2)^2(x-4)$.***

Example 12: Sketch It

$$f(x)=-2(x-3)^2(x+1)$$

Question

CheckIt will ask you to sketch this. Reading from left to right, which description of the graph is right?

A. Starts low, goes through the axis at −1, turns at 3, ends high
B. Starts high, turns at −1, goes through the axis at 3, ends low
C. Starts high, goes through the axis at −1, turns at 3, ends low
D. Starts high, goes through the axis at 1, turns at −3, ends low

Answer and reasoning

Degree $2+1=3$ with a negative leading coefficient: up on the left, down on the right. Coming down from the upper left, the graph crosses at $x=-1$ (multiplicity 1), dips to a low point, passes through the y-intercept $-2(-3)^2(1)=-18$, rises to touch the axis at $x=3$ (multiplicity 2), and turns back down. Starting low is the positive version, turning at −1 swaps the multiplicities, and 1 and −3 read the signs as written. **Starts high, crosses at −1, touches at 3, ends low.** To sketch it, plot $(-1,0)$, $(3,0)$, and $(0,-18)$, then connect them this way.

Example 13: Slip 1

$$f(x)=3(x+4)^2(x-1)$$

Question

A student says the zeros are 4, with multiplicity 2, and −1, with multiplicity 1. What is the mistake, if any?

A. There is no mistake.
B. The multiplicities are swapped: 4 has 1 and −1 has 2.
C. The signs are flipped: the zeros are −4 and 1.
D. 3 is also a zero, with multiplicity 1.

Answer and reasoning

Set each factor equal to 0: $x+4=0$ gives $x=-4$, and $x-1=0$ gives $x=1$. The multiplicities were right. **The zeros are −4, with multiplicity 2, and 1, with multiplicity 1.**

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Example 14: Slip 2

$$f(0)=2(0-1)^2(0+3)$$
$$=2\cdot 1\cdot 3$$
$$=6$$

Question

For f(x) = 2(x − 1)²(x + 3), a student found the y-intercept (0, 6) with this work. What is the mistake, if any?

A. There is no mistake.
B. (−1)² is −1, so the intercept is (0, −6).
C. The y-intercept is where y = 0, so x = 1.
D. It should be (0, 2), from the leading coefficient.

Answer and reasoning

The y-intercept is where $x=0$. $(0-1)^2=(-1)^2=1$, and $2\cdot 1\cdot 3=6$. **There is no mistake.**

Example 15: Slip 3

$$f(x)=(x+2)^2(x-3)$$

Question

A student sketched the graph crossing the x-axis at x = −2 and at x = 3. What is the mistake, if any?

A. There is no mistake.
B. At −2 the multiplicity is 2, so it turns there.
C. It should turn around at 3, not at −2.
D. The zeros should be at 2 and −3, not at −2 and 3.

Answer and reasoning

$(x+2)$ is squared, so $x=-2$ has even multiplicity and the graph touches the x-axis and turns around there. At $x=3$, multiplicity 1, it does cross. **Touch at −2, cross at 3.**

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Example 16: Slip 4

$$f(x)=-2(x-3)^2(x+1)$$

Question

A student says both ends go down, because the leading coefficient is negative. What is the mistake, if any?

A. There is no mistake.
B. Both ends go up, because 2 squared is positive.
C. The left end goes down and the right end goes up.
D. The degree is odd, so the ends go opposite ways.

Answer and reasoning

The degree is $2+1=3$, so the ends go opposite ways. With a negative leading coefficient, the graph goes up on the left and down on the right. "Both ends down" is right only for even degree. **Up on the left, down on the right.**

Example 17: Slip 5

$$f(0) = -4(0-2)^2(0+5)$$
$$= -4 \cdot (-2) \cdot 5$$
$$= 40$$

Question

For f(x) = −4(x − 2)²(x + 5), a student found the y-intercept with this work. What is the mistake, if any?

A. There is no mistake.
B. The square was skipped: (−2)² = 4, so it is (0, −80).
C. The y-intercept should be (0, −4), the leading coefficient.
D. It should be (0, 80), since squares are positive.

Answer and reasoning

$(0-2)^2 = (-2)^2 = 4$, not -2. So $f(0) = -4 \cdot 4 \cdot 5 = -80$. The square makes that factor positive, and the sign comes from the −4. **The *y*-intercept is *(0,−80)*.**